Zhefeng Huang

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RESEARCH INTERESTS

Robot learning, world models, vision-language-action models, task and motion planning, robot manipulation, and medical and surgical robotics.

EDUCATION

Georgia Institute of Technology **Aug 2022 - Present**
Ph.D. in Robotics, Atlanta, GA, USA
- GPA: 4.00/4.00; passed the Robotics Ph.D. Qualifying Examination.

Tsinghua University **Sep 2018 - Jun 2022**
B.S. in Mechanical Engineering (Elite Program), Beijing, China
- Major GPA: 3.83/4.00; Minor in Computer Technology and Applications, GPA: 3.94/4.00.

RESEARCH EXPERIENCE

World-Model-Augmented VLA for Long-Horizon Surgical Manipulation **Jun 2026 - Present**
Georgia Institute of Technology, Atlanta, GA
- Developing vision-language-action policies augmented with future-predictive latent world models for long-horizon surgical manipulation.
- Learning predictive state representations from mixed-quality expert and non-expert interaction data.

Flow-Matching World Model for Robot Manipulation Failure Detection **Feb 2026 - Jul 2026**
Georgia Institute of Technology, Atlanta, GA
- Developed an action-conditioned flow-matching latent world model to detect low-level execution failures using only successful training data.
- Compared expected and observed visual dynamics through inverse latent transport and calibrated operational thresholds with conformal prediction.
- Built simulation and real-world da Vinci Research Kit pipelines for data collection, imitation learning, and evaluation.

Self-Reconfiguration Planning for Continuum Modular Robot System **Apr 2025 - Jan 2026**
Georgia Institute of Technology, Atlanta, GA
- Developed geometric and topological representations of planar continuum modular robots using combinatorial topology.
- Formulated self-reconfiguration as task and motion planning over configuration manifolds; proposed HEART-MCTS and integrated Atlas-RRT*.
- Developed a graph neural network representation of modular robot morphology.

Mobile Nursing Robot in an Intensive Care Unit **Apr 2024 - Oct 2024**
Georgia Institute of Technology / Emory Decatur Hospital, Atlanta, GA
- Developed a customized teleoperation interface for a Stretch 2 mobile manipulator and conducted teleoperated demonstrations in an ICU ward.
- Conducted a human-robot interaction study of nurses' perceptions and acceptance of mobile nursing robots.

Ultrasound-Guided Robotic Needle Insertion **Sep 2023 - Jan 2024**
Georgia Institute of Technology, Atlanta, GA
- Developed a dual-arm robotic needle insertion system using a Franka Research 3 arm and ultrasound guidance.
- Built an XY motion platform to emulate respiratory motion and performed insertions during relative target quiescence.

Development of a Body-Mounted MR-Conditional Medical Robot **Jan 2023 - Sep 2023**
Georgia Institute of Technology, Atlanta, GA
- Designed, fabricated, and integrated a four-degree-of-freedom body-mounted MR-conditional robot with a two-layer positioning structure and pneumatic actuation.
- Developed the mechatronic control system and evaluated robot kinematics, workspace, and targeting performance.

Geometric Modeling and Optimization of Concentric Tube Robots **Sep 2022 - Jan 2023**
Georgia Institute of Technology, Atlanta, GA
- Developed parameterized helical models and optimization methods for concentric-tube robot design under anatomical constraints for intracerebral hemorrhage evacuation.

PROFESSIONAL EXPERIENCE

Mechanical Engineering Intern **Jun 2021 - Aug 2021**
Ericsson, Beijing, China
- Developed a Microsoft Excel add-in to streamline tolerance-analysis workflows; the tool was adopted by the mechanical design team.
- Investigated airflow-induced vibration and noise of heat-sink fins in outdoor telecom base stations through coupled fluid-structure-acoustic simulation and wind-tunnel testing; proposed design modifications to reduce wind noise.

SELECTED TECHNICAL PROJECTS

Local MCTS Motion Planning - *Robotics Ph.D. Qualifying Examination Project* **May 2024**
- Implemented a Monte Carlo tree search local planner for a non-holonomic mobile robot navigating among passively deformable obstacles.
- Enabled the robot to push an obstacle within its deformation limit, proceed when a path opened, or retreat and replan when continued interaction was infeasible.

Procedural Trees and Creatures – *Course Project, CS 8803: Procedural Content Generation* **Fall 2024**
- Implemented classic procedural content generation algorithms in C# using Unity.
- Created randomized tree-growth systems with controllable branching, tropism, leaf placement, age-dependent growth, and obstacle-aware generation.
- Generated continuous creature meshes using Marching Cubes with shared vertices for smooth normals and color transitions, together with randomized morphology and accessories.

TECHNICAL SKILLS

Robot Learning & AI: Imitation learning, world models, vision-language-action models, flow matching, computer vision
Planning & Robotics: Task and motion planning, MCTS, sampling-based motion planning, robot kinematics, manipulation
Programming & Frameworks: Python, C++, C, MATLAB, PyTorch, ROS 2, Isaac Sim, OpenCV, MuJoCo, Unity
Robots & Engineering: da Vinci Research Kit, Franka Research 3, Stretch 2, SolidWorks, ANSYS, COMSOL

PUBLICATIONS

Preprint and Manuscript Under Review

1. **Zhefeng Huang**, Yilin Cai, Ankit Patel, Mohammad Hajiha, Brendan Browne, and Yue Chen. "Failure Detection for Surgical Robot Imitation Policies via Flow-Matching World Modeling." *arXiv preprint arXiv:2607.27511*, 2026. Submitted to IEEE Robotics and Automation Letters (RA-L).

Peer-Reviewed Journal Articles

2. Yilin Cai*, **Zhefeng Huang***, Yifan Wang, Haokai Xu, and Yue Chen. "Evolutionary Diversification via Modular Compliance for Self-Reconfigurable Continuum Robots." *Science Advances* 12, eaeg9191, 2026. *Equal contribution.
3. **Zhefeng Huang**, Anthony L. Gunderman, Samuel E. Wilcox, Saikat Sengupta, Jay Shah, Aiming Lu, David Woodrum, and Yue Chen. "Body-Mounted MR-Conditional Robot for Minimally Invasive Liver Intervention." *Annals of Biomedical Engineering* 52(8), 2024, pp. 2065-2075.
4. **Zhefeng Huang**, Hussain Alkhars, Anthony Gunderman, Dimitri Sigounas, Kevin Cleary, and Yue Chen. "Optimal Concentric Tube Robot Design for Safe Intracerebral Hemorrhage Removal." *Journal of Mechanisms and Robotics* 16(8), 2024.
5. Anthony L. Gunderman, Saikat Sengupta, **Zhefeng Huang**, Dimitri Sigounas, Chima Oluigbo, Isuru S. Godage, Kevin Cleary, and Yue Chen. "Towards MR-Guided Robotic Intracerebral Hemorrhage Evacuation: Aiming Device Design and Ex Vivo Ovine Head Trial." *IEEE Transactions on Medical Robotics and Bionics*, 2024.
6. Samuel E. Wilcox, **Zhefeng Huang**, Jay Shah, Xiaofeng Yang, and Yue Chen. "Respiration-Induced Organ Motion Compensation: A Review." *Annals of Biomedical Engineering* 53(2), 2025, pp. 271-283.

Conference Proceedings

7. Joseph Sommer, Anthony L. Gunderman, Saikat Sengupta, **Zhefeng Huang**, Dimitri Sigounas, Kevin Cleary, and Yue Chen. "Concentric Tube Robot-Based Intracerebral Hemorrhage Evacuation in Ex Vivo Sheep Head: A Comparative Study." In *2024 International Symposium on Medical Robotics (ISMR)*, pp. 1-7, IEEE, 2024.

AWARDS AND HONORS

Outstanding Graduate, Tsinghua University **Jun 2022**
Tang Lixin Scholarship (Comprehensive Excellence), Tsinghua University **Sep 2021**
Academic Excellence Scholarship, Tsinghua University **Sep 2020**

LEADERSHIP AND ATHLETICS

Vice Captain, Tsinghua University Student Track and Field Team **2020 - 2021**
Three-Time Men's Group A High Jump Champion, 62nd-64th Ma John Cup **2019 - 2021**